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Van der Waals quaternary oxides for tunable low-loss anisotropic polaritonics

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The discovery of ultraconfined polaritons with extreme anisotropy in a number of van der Waals (vdW) materials has unlocked new prospects for nanophotonic and optoelectronic applications. However, the range of suitable materials for specific applications remains limited. Here we introduce tellurite molybdenum quaternary oxides—which possess non-centrosymmetric crystal structures and extraordinary nonlinear optical properties—as a highly promising vdW family of materials for tunable low-loss anisotropic polaritonics. By employing chemical flux growth and exfoliation techniques, we successfully fabricate high-quality vdW layers of various compounds, including $MgTeMoO_6$, $ZnTeMoO_6$, $MnTeMoO_6$ and $CdTeMoO_6$. We show that these quaternary vdW oxides possess two distinct types of in-plane anisotropic polaritons: slab-confined and edge-confined modes. By leveraging metal cation substitutions, we establish a systematic strategy to finely tune the in-plane polariton propagation, resulting in the selective emergence of circular, elliptical or hyperbolic polariton dispersion, accompanied by ultraslow group velocities (0.0003*c*) and long lifetimes (5 ps). Moreover, Reststrahlen bands of these quaternary oxides naturally overlap that of α-MoO_3, providing opportunities for integration. As an example, we demonstrate that combining α-MoO_3 (an in-plane hyperbolic material) with $CdTeMoO_6$ (an in-plane isotropic material) in a heterostructure facilitates collimated, diffractionless polariton propagation. Quaternary oxides expand the family of anisotropic vdW polaritons considerably, and with it, the range of nanophotonics applications that can be envisioned.

The tellurite molybdenum/tungsten (A–Te–Mo/W–O) quaternary oxides are a highly intriguing class of nonlinear optical crystals with remarkable multifunctional properties[1–5]. Their diverse compositions—including $ATeMoO_6$ (A = Mg, Zn, Mn, Cd, Co, etc.), $BaTe_2MO_9$ (M = Mo, W), $Cs_2TeM_3O_{12}$ (M = Mo, W) and others—typically possess non-centrosymmetric crystal structures, which give rise to their nonlinear optic, piezoelectric and electro-optical characteristics. Their applications are broad-ranging, spanning from strong second-harmonic generation[1,3], piezoelectric resonators[2,6] and acousto-optical modulation[7], among others[8]. In addition to nonlinear optical features, these oxides also possess other interesting properties. For instance, the strong covalent bonding of the Mo/W groups leads

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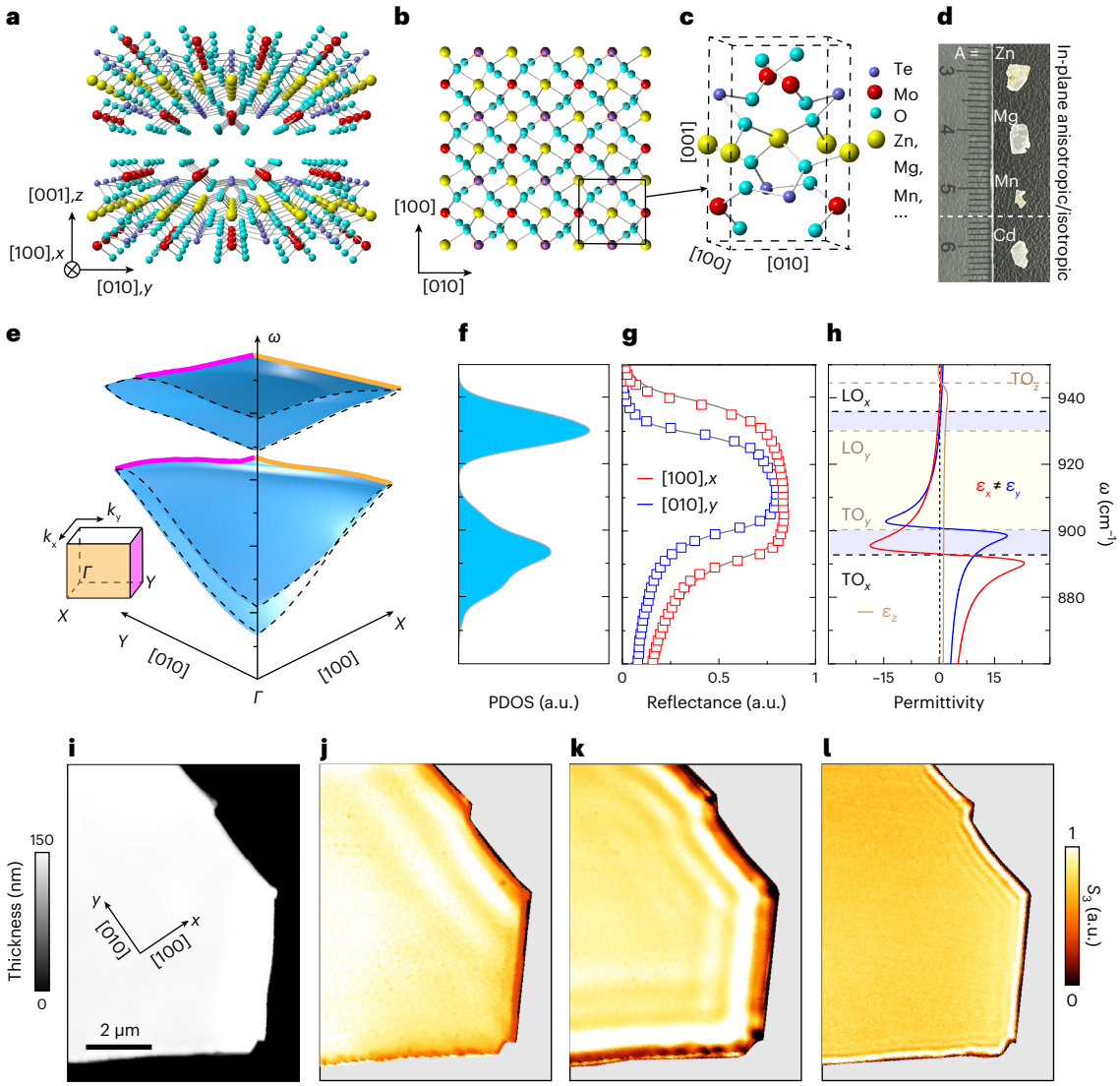

Fig. 1 | Optical properties of vdW quaternary oxides. a,**b**, Illustration of the typical orthorhombic lattice structure of $ATeMoO_6$ in side view (**a**) and top view (**b**). **c**, Schematic of the unit cell. For $ZnTeMoO_6$, the lattice constants along the principal [100], [010] and [001] axes are indicated as $a = 0.504$, $b = 0.526$ and $c = 0.892$ nm, respectively. These compounds typically crystallize in the orthorhombic space group[3] $P2_12_12$. However, $CdTeMoO_6$ has a tetragonal crystal structure[11], belonging to the space group $P42_1m$. **d**, Optical image of the grown crystals of $ATeMoO_6$. A grid of the ruler represents 1 mm. **e**, Topological phonon band structures of $ZnTeMoO_6$ in the Γ–XY plane obtained through first-principles calculations. The symmetry paths are represented by dashed black lines, and by magenta and orange lines. The magenta and orange lines located on the $k_x = \pi$ and $k_y = \pi$ plane indicate two-nodal surface phonons in $ZnTeMoO_6$. **f**, Calculated PDOS within the phonon band. **g**, Polarized FTIR reflection spectra of $ZnTeMoO_6$. Solid lines indicate the calculated infrared reflection spectra of $ZnTeMoO_6$; symbols represent the experimental data. **h**, The real parts of the permittivity (continuous lines) along the principal x, y and z axes (represented by red, blue and yellow lines). The light yellow and light grey regions indicate the elliptical and hyperbolic bands, respectively. **i**, AFM topography image of a 140-nm-thick $ZnTeMoO_6$ flake. **j**–**l**, Near-field optical amplitude images of the flake at frequencies of 900 cm^{-1} (**j**), 906 cm^{-1} (**k**) and 920 cm^{-1} (**l**). To better observe the fringes, the SiO_2 substrate is masked by grey.

to intense and narrow phonon vibration modes, enabling Raman laser output capable of generating THz waves through frequency differences[9]. Theoretical calculations have also predicted the existence of topological phonons in these materials[10], such as in $ZnTeMoO_6$, offering intriguing possibilities for investigating exotic vibrational phenomena and light–matter interactions.

While the focus has traditionally been on producing large, high-quality single crystals for technological applications[11], certain subsets of the quaternary oxides, such as the $ATeMoO_6$ compounds, exhibit a propensity for layered structures due to weak interlayer interactions[3]. These layered crystals, held together by van der Waals (vdW) forces, present great potential for two-dimensional (2D) electronics and optoelectronics[12–14].

In this article, we have successfully obtained high-quality vdW layers of various compounds, including $MgTeMoO_6$, $ZnTeMoO_6$, $MnTeMoO_6$ and $CdTeMoO_6$, through chemical growth and mechanical exfoliation into 2D layered materials (Methods). We present experimental results uncovering phonon polaritons (PhPs) within this class of vdW materials, where the hybridization of photons and optical phonons gives rise to these quasiparticles. Through advanced infrared nanoimaging, we have directly visualized in-plane anisotropic PhPs propagating with hyperbolic, elliptical or circular dispersion within thin slabs of $ATeMoO_6$ and discovered the presence of in-plane anisotropic edge-confined PhPs at the sidewalls of these vdW materials. By tuning optical phonons and their associated PhPs through facile substitution of metal cations in quaternary oxides, we introduce a

diverse family of polaritonic vdW materials supporting low-loss tunable anisotropic polaritons.

The structure and properties of vdW quaternary oxides

The atomic structure of $ATeMoO_6$ (A = Mg, Zn, Mn, Co, etc.) compounds consists of layered structures bound by weak vdW forces (Fig. 1a). As an example, for $ZnTeMoO_6$, ZnO_6 octahedra are connected via corner sharing, forming 2D layers that alternate with MoO_4 tetrahedra and TeO_4 polyhedra stacked in an up–down configuration (Fig. 1a,b). This arrangement leads to unequal lattice constants (Fig. 1c), a and b, resulting in structural in-plane anisotropy. However, $CdTeMoO_6$ exhibits in-plane isotropy because regular CdO_4 tetrahedra occupy the positions of distorted MoO_6 octahedra in its crystal structure[11].

The optical image (Fig. 1d) shows the successfully grown crystals, while the mechanically exfoliated thin flakes exhibit irregular shapes, in contrast to the ribbon-like structures typically observed in orthorhombic crystals like α-MoO_3 (refs.15–17). The in-plane crystallographic orientations (x = [100], y = [010]) of the flakes can be precisely discriminated (Supplementary Note 1). Recent advancements in the understanding of $ATeMoO_6$ compounds, particularly $ZnTeMoO_6$, have highlighted the influence of Te–O and Mo–O bonds on the generation of exotic phonon modes[10]. For example, theoretical predictions have revealed the intriguing presence of topological nodal-surface phonons in $ZnTeMoO_6$, which possess great potential, including freedom from Fermi-level restrictions and exceptional thermal stability due to minimal scattering during transport processes[18].

We conducted first-principles calculations to illustrate the predicted nodal-surface phonons within the phonon band structures of $ZnTeMoO_6$. Figure 1e showcases the calculated band structures. The phonon density of states (PDOS) depicted in Fig. 1f exhibits two distinct regimes or peaks, corresponding to the contributions of distinct phonon modes. This analysis reveals the presence of two flat phonon bands (magenta and orange lines) close to the frequencies ω = 930 and 950 cm^{-1}. These flat bands unequivocally occupy the Γ–XY plane within this frequency range, indicating the existence of ideal nodal surfaces along this plane[10] defined by $k_x = \pi$ and $k_y = \pi$ in the inset in Fig. 1e. Furthermore, the observed inherent asymmetry in the phonon bands along the [100] and [010] directions in $ZnTeMoO_6$ suggest that the material exhibits distinct optical properties along different crystallographic directions, potentially leading to anisotropic optical phenomena and functionalities.

To experimentally investigate the optical anisotropy of $ZnTeMoO_6$, we conducted polarization-resolved far-field Fourier transform infrared (FTIR) spectroscopy (see Methods for details). Figure 1g presents the reflection spectra measured parallel to the x and y directions, revealing the pronounced reflectivity in the Reststrahlen band (RB) frequency range from ω = 860 to 950 cm^{-1}. To extract the permittivity components ($\varepsilon_x, \varepsilon_y, \varepsilon_z$) along the three crystallographic directions, we fitted the reflection spectra using Lorenz oscillator models and finely adjusted the parameters with near-field experimental data (Supplementary Note 2). The obtained fitting shows excellent agreement with the experimental data (see Fig. 3g). As shown in Fig. 1h, the signs of the two lateral components ε_x and ε_y segregate the frequency range into different spectral bands, denoted by the yellow ($\varepsilon_x\varepsilon_y > 0, \varepsilon_x, \varepsilon_y < 0$) and light grey ($\varepsilon_x\varepsilon_y < 0$) regions. Based on previous studies on anisotropic polaritons[15,16,19,20], such anisotropic permittivity suggests that $ZnTeMoO_6$ can support elliptical and hyperbolic polaritons in respective frequency ranges.

We employed scattering-type scanning near-field optical microscopy (s-SNOM) based on an atomic force microscope (AFM) to directly visualize the anisotropic polaritons[21–32]. Figure 1i displays the topography of a 140-nm-thick $ZnTeMoO_6$ flake. Experimental near-field amplitude images at three different frequencies (Fig. 1j–l) revealed periodic bright and dark fringes within the flake. These fringes originate from the interference of polaritons launched by the probing tip, propagating along the flake and reflecting at its edges. The spacing between fringes corresponds to half the polariton wavelength in the respective directions[15,21,33,34].

Analysing near-field images at ω = 900 cm^{-1}, we find that the fringes exclusively appeared along the [100] direction while being absent along other directions. This finding could signify the presence of in-plane hyperbolic-polariton responses, as predicted from the anisotropic permittivity (Fig. 1g). In contrast, at ω = 906 cm^{-1}, we observed the fringes running parallel to all edges of the flake. They displayed distinct periodicities and quantities depending on the propagating direction (Supplementary Fig. 6), consistent with the anticipated elliptical polariton responses in Fig. 1g. At ω = 920 cm^{-1}, the fringes become more tightly confined, indicating the increased confinement of polariton fields.

Intriguingly, we observed periodic bright/black dots representing near-field signals along the flake edges in the near-field image obtained at ω = 906 cm^{-1}. This strongly suggests the presence of edge-confined polariton modes[35,36] in $ZnTeMoO_6$. We found the near fields of these edge modes to extend over 1,000 nm into the flake (Supplementary Fig. 6), making them less affected by small changes in the edge shape. The fringes associated with the edge modes exhibit a smaller oscillation period and consequently a smaller wavelength ($\lambda_{edge} < \lambda_{slab}$) (Supplementary Fig. 6), indicating stronger confinement of polariton fields in the edge modes.

Real-space nanoimaging of anisotropic and tunable polaritons

Experiments on other flakes verify the coexistence of both slab- and edge-confined modes (Fig. 2a–d). We find that the edge-confined polaritonic patterns show different periodicities along different edge directions. They are also sensitive to the operation frequency, indicating sharp frequency dispersion, and thus slow group velocities. The near-field profiles extracted along two orthogonal edges (parallel to the [001] and [010] directions; Fig. 2c) showcase the anisotropic nature of edge modes. The near-field profiles (Fig. 2e and Supplementary Fig. 7) show distinct periods of these edge modes, thus revealing their in-plane anisotropy. In-plane isotropic edge modes have been observed in hexagonal boron nitride (hBN)[35,36], but not in other anisotropic vdW materials[15,16,19] (α-MoO_3, α-V_2O_5, etc.; Supplementary Table 6). This stems from the critical condition of supporting anisotropic edge modes ($\varepsilon_x, \varepsilon_y < 0, \varepsilon_x \neq \varepsilon_y$ and $\varepsilon_z > 0$)[35,36] that only appear within a narrow frequency range, making them challenging to observe. For α-MoO_3 and α-V_2O_5, the required frequency range is below 850 cm^{-1} (shown numerically in Supplementary Note 3), which cannot be easily reached in experiments.

We further conducted numerical simulations to validate the observed in-plane anisotropic edge modes in $ZnTeMoO_6$ flakes. Figure 2f presents the simulated near-field distributions for two edges ([100] and [010] directions) of $ZnTeMoO_6$. Simulations (left panels) confirm strong confinement of the polariton field at the edges for both directions, with a rapid decay when moving away from the edge. The polariton field along the [100] direction penetrates deeper into the slab, resulting in a faster decay in the surrounding environment compared with the [010] direction. Consequently, the wavelength of the edge modes along the [100] direction is shorter than the other direction (see right panels of the simulations).

Similar to prior studies[21,37], anisotropic slab modes in $ZnTeMoO_6$ arise from the volume-confined hyperbolic polaritons (illustrated in Fig. 2g, exhibiting a 3D (yellow plot) open hyperboloid isofrequency contour (IFC) in the k-space) propagating within the bulk of $ZnTeMoO_6$. In stark contrast, in-plane anisotropic edge modes originate from the surface-confined hyperbolic polaritons[35,36] (displaying 2D hyperbolic IFCs; red curves in Fig. 2g) that exist at the sidewalls of the material.

For a thin layer with a finite thickness d, the volume hyperbolic polaritons are bounded by the top and bottom slab interfaces,

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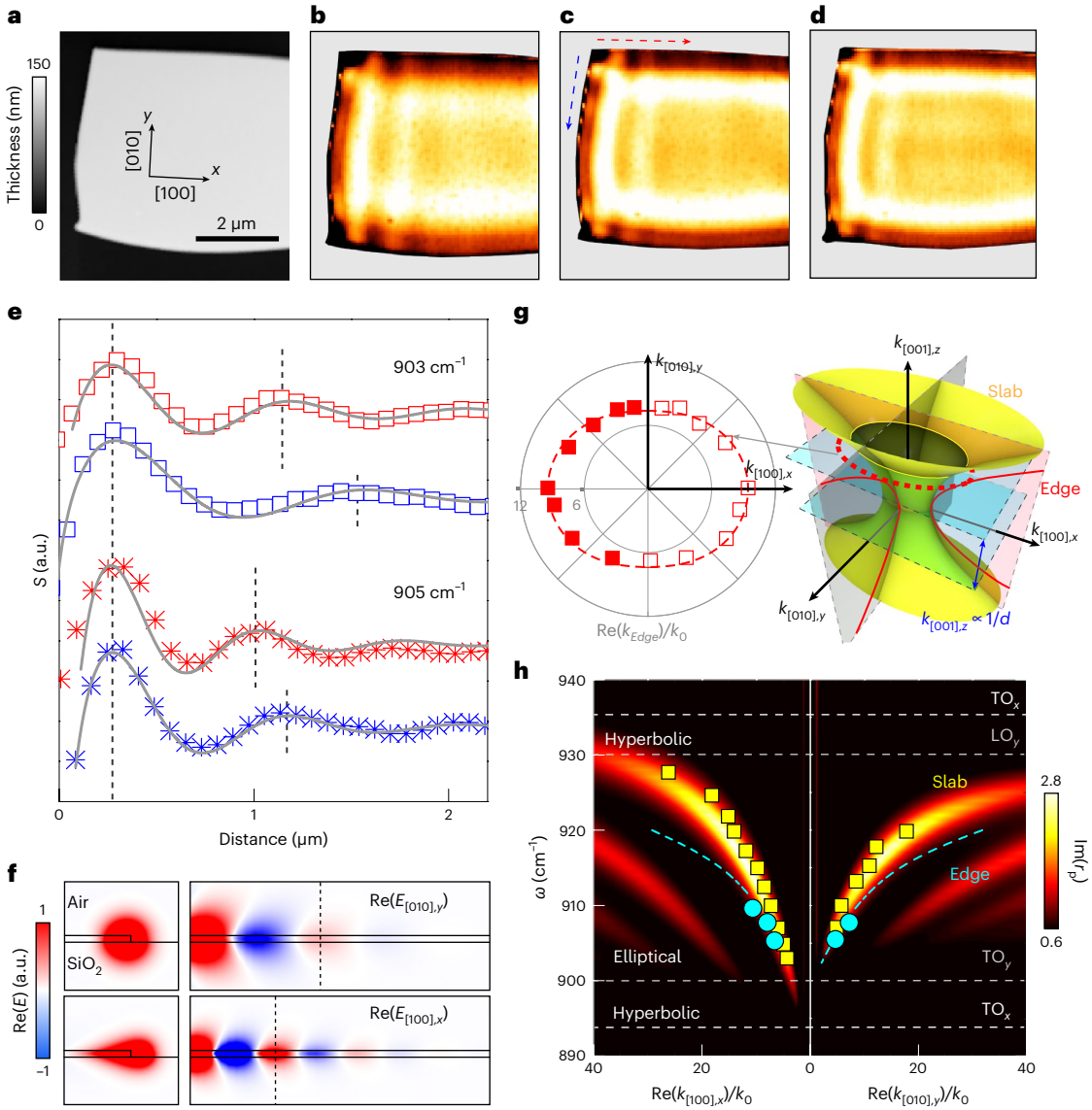

Fig. 2 | Observation of in-plane anisotropic edge polaritons. a, AFM topography image of a 142-nm-thick $ZnTeMoO_6$ flake. **b**–**d**, Near-field amplitude images of the $ZnTeMoO_6$ flake at frequencies of 903 cm^{-1} (**b**), 905 cm^{-1} (**c**) and 906 cm^{-1} (**d**). The blue and red dashed arrows represent different edges of the flake. **e**, Detailed profiles taken from the flake edges as indicated in **b**,**c**. Symbols, experimental data; grey lines, fittings. The vertical dashed lines represent the peaks of intensity. **f**, Simulated near-field distribution of the edge modes along the two characteristic directions. The top panel shows the results for the [010] direction (left, x–z plane; right, y–z plane). The bottom panel shows the cases for the [100] direction (left, y–z plane; right, x–z plane). The results were simulated for a slab of $ZnTeMoO_6$ with a thickness of 80 nm at $\omega = 902$ cm^{-1}. **g**, Left: the in-plane wavevector confinement of edge modes at $\omega = 907$ cm^{-1}, where the hollow squares (raw data) represent experimental results obtained from several flakes with similar thicknesses. The solid squares arise from the mirror images of the hollow squares along the vertical line. The dashed red line shows the simulated results. Right: the origin of the anisotropic edge modes in k-space. Yellow surface, 3D hyperboloid IFC of volume polaritons in $ZnTeMoO_6$; solid red lines, 2D hyperbolic IFCs of surface polaritons at $ZnTeMoO_6$ sidewalls along different lateral directions. The in-plane IFCs of the slab and edge modes (the solid yellow line and the dashed red line, respectively) can be conceptualized as cross sections of the 3D IFCs of volume polaritons or the 2D IFCs of surface polaritons, cut at a specific position of $k_{[001],z}k_{[001],z}/k_0 = (m+1)\lambda_0/2d$, $m = 0, 1, \ldots$ ($m = 0$ represents the fundamental edge modes). **h**, Colour map illustrating the imaginary part of the Fresnel reflection coefficient, Im(r_p), for the calculated dispersion for slab modes in a 190-nm-thick slab of $ZnTeMoO_6$ along two directions, [100] and [010]. Dashed blue lines, calculated dispersion for edge modes; symbols, experimental data.

leading to the presence of slab modes that are guided along the layer. The surface hyperbolic polaritons, on the other hand, are confined by the top and bottom corners of the sidewalls, resulting in the existence of edge modes. The in-plane IFCs of the slab and edge modes can be conceptualized as cross sections of the 3D hyperboloid dispersion of volume polaritons or the 2D hyperbolic dispersion of surface polaritons, cut at a specific position of $k_{[001],z} \propto 1/d$ (refs. 21,37–40). For $ZnTeMoO_6$ with $\varepsilon_x \neq \varepsilon_y$, we obtain two elliptical IFCs (depicted by solid yellow and dashed red plots in Fig. 2g) for both slab and edge modes, which provide a comprehensive explanation for the observed in-plane anisotropic responses of both modes in experiments.

To experimentally validate elliptical in-plane IFCs of edge modes, we exfoliated a series of $ZnTeMoO_6$ flakes with a similar thickness of 140 nm, each possessing different edges oriented along distinct in-plane directions. Subsequently, we performed near-field imaging of edge modes on those edges (Supplementary Fig. 8). The experimental edge-mode wavevectors obtained in different directions (red square

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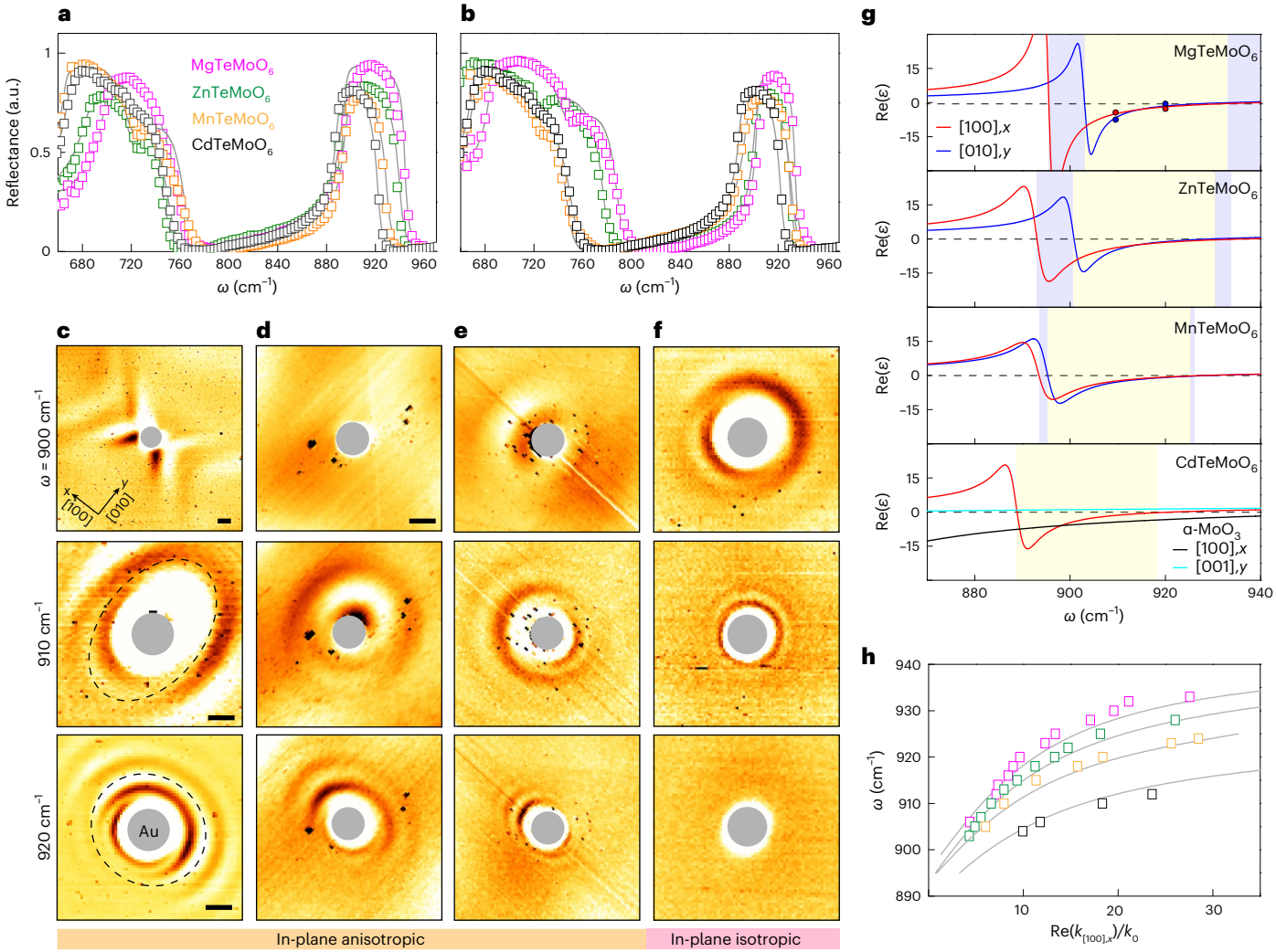

Fig. 3 | Tailoring polariton responses by atomic substitution. a,**b**, Polarized FTIR reflection spectra of bulk $ATeMoO_6$ (A = Zn, Mg, Mn, Cd) along the [100] (**a**) and [010] (**b**) directions. Symbols, experimental data; grey lines, fittings. The reflection curves show a noticeable shift, indicating the shift of RBs and optical phonons in these materials. **c**–**f**, Near-field images of polaritons launched by gold antennas fabricated on $MgTeMoO_6$ (**c**), $ZnTeMoO_6$ (**d**), $MnTeMoO_6$ (**e**) and $CdTeMoO_6$ (**f**). Grey disks mask the gold antennas (diameter, ~1.6 μm). Black dashed lines denote the elliptical polaritonic wavefront patterns. Scale bar, 1 μm. The thicknesses for $MgTeMoO_6$, $ZnTeMoO_6$, $MnTeMoO_6$ and $CdTeMoO_6$ are 195 nm, 238 nm, 202 nm and 180 nm, respectively. **g**, Real part of the permittivity of $ATeMoO_6$ along the principal [100], [010] directions (red and blue solid lines, respectively). The black and bright blue solid lines indicate the real part of the permittivity of α-MoO_3 along the principal [100], [001] directions. The light yellow and light grey regions indicate the elliptical and hyperbolic bands, respectively. The red and blue symbols denote the calculated permittivity of $MgTeMoO_6$ at the corresponding frequencies. $MgTeMoO_6$ at $\omega = 900\ cm^{-1}$ supports in-plane hyperbolic polaritons due to $\varepsilon_{x,[100]}\varepsilon_{y,[010]} < 0$. Meanwhile, the permittivity trends, with $|\varepsilon_{x,[100]}| < |\varepsilon_{y,[010]}|$ at $\omega = 910\ cm^{-1}$ and $|\varepsilon_{x,[100]}| > |\varepsilon_{y,[010]}|$ at $\omega = 920\ cm^{-1}$, cause the change in the major axis of the elliptical wavefronts. **h**, Dispersion of polaritons along the [100] direction in these materials. Square symbols are experimental data extracted along the [100] direction in **c**,**e**,**f**. The symbols for $ZnTeMoO_6$ are extracted from Fig. 2h for the approximate consistency of the thickness. The confinement ($Re(k_{[100]})/k_0$) changes dramatically due to atomic substitution. Grey lines are the analytically calculated polariton dispersion.

symbols in Fig. 2g) agree well with the calculated elliptical dispersion (dashed red line), confirming the in-plane anisotropy of edge modes.

To further understand anisotropic polariton responses, we employed theoretical calculations to precisely determine the frequency dispersion of slab and edge modes along the [100] and [010] directions. Figure 2h presents a false-colour map with the discernible peaks corresponding to the slab modes[21] for the $ZnTeMoO_6/SiO_2$ sample system. The slab modes exhibit distinct dispersion branches in each direction: a lower branch ($890 < \omega < 940\ cm^{-1}$) with positive dispersion and an upper branch ($940 < \omega < 960\ cm^{-1}$) with negative dispersion[15,21,38,41] (Supplementary Fig. 9). The dispersion branches between the two directions reveal the in-plane anisotropy. Specifically, in the [100] direction, the lower branch extends up to $\omega = 930\ cm^{-1}$, while in the [010] direction, it remains below $930\ cm^{-1}$. This disparity results in different frequency ranges for the in-plane hyperbolic or elliptical dispersion of the slab modes. The dashed blue lines represent the calculated dispersion of edge modes ($900–930\ cm^{-1}$), with larger wavevectors and stronger confinement compared to the slab modes at the given frequency. Our experimental dispersion data (symbols in Fig. 2h; Supplementary Fig. 10) for both slab and edge modes demonstrate excellent agreement with the theoretical dispersion, verifying the validity of our experimental observations.

Having established the anisotropic polariton responses in $ZnTeMoO_6$, we now focus on the substantial tunability of polariton properties within the $ATeMoO_6$ family, facilitated by the flexible substitution of metal cations during synthesis[42]. Despite the diverse A-atoms

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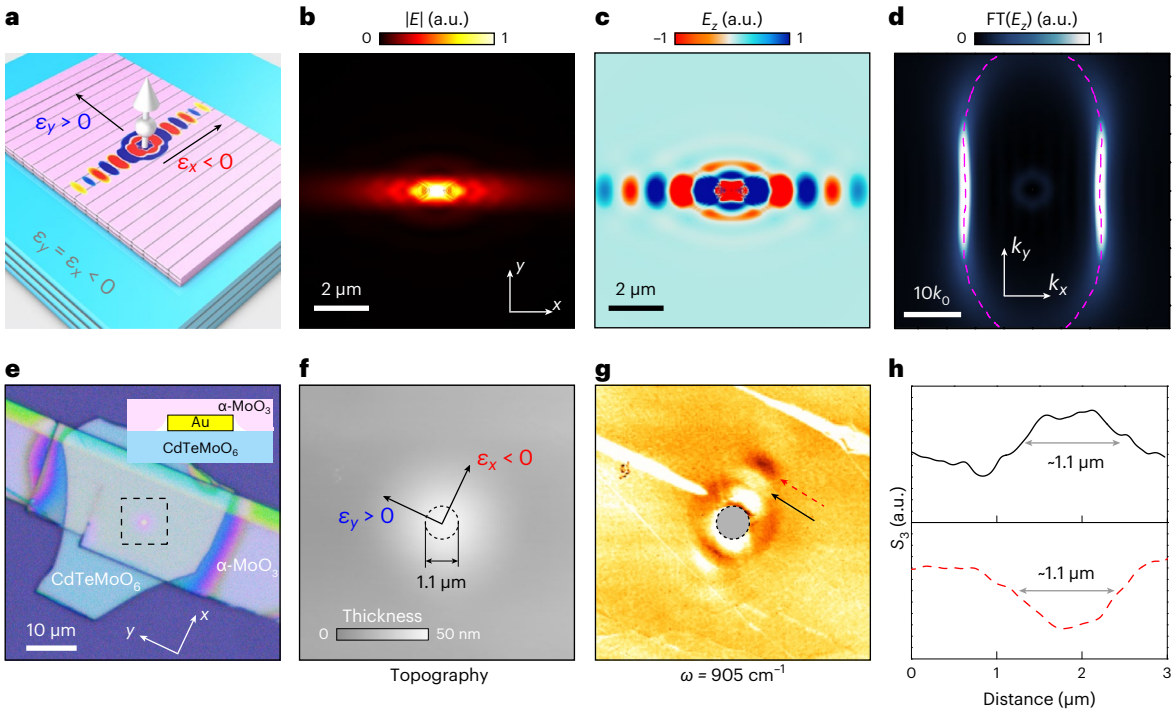

Fig. 4 | Diffractionless nanolight canalization by the heterostructure without twist. a, Dipole launching of polariton canalization in the heterostructure. Top layer, in-plane hyperbolic materials; bottom layer, in-plane isotropic materials. We chose α-MoO_3 as the upper layer due to its lower losses, and $CdTeMoO_6$ as the bottom layer in the simulation. **b**, Simulated magnitude (|*E*|) of the near-field distribution at 905 cm^{-1} above α-MoO_3/$CdTeMoO_6$ heterostructure. Thickness: α-MoO_3, 150 nm; $CdTeMoO_6$, 200 nm. **c**, Simulated near-field distribution (E_z) at 905 cm^{-1}. **d**, Absolute value of the Fourier transform (FT) of **c**. The magenta dashed lines are the IFCs obtained using the transfer matrix method. **e**, Optical image of α-MoO_3/$CdTeMoO_6$ heterostructure. Thickness: α-MoO_3, 170 nm; $CdTeMoO_6$, 220 nm. A gold disk (diameter, ~1.1 μm) is positioned between the top and bottom layers. **f**, AFM topography of this heterostructure obtained from the black dashed square in **e**. The black dashed circle in **f**,**g** denotes the gold disk. **g**, Near-field image of polariton canalization (diffractionless propagation) of the heterostructure at 905 cm^{-1}. The bright white regions result from the crack of the α-MoO_3 flake from **e**. The red and black arrows denote the peaks and troughs of intensity shown in **h**. **h**, The line profiles crossing the polaritonic patterns taken through the arrows in **g**. The half-width at half-maximum is ~1.1 μm.

(Mg, Zn, Mn, Cd, etc.) originating from different rows of the periodic table, they consistently maintain a valence state of +2, ensuring the equilibrium of electron gain and loss. This atomic substitution process introduces variations in the mass of the A-atom, consequently influencing the lattice vibrations and phonons associated with neighbouring bonds.

Figure 3a,b presents the polarized reflection spectra of four various compounds along the [100] and [010] directions, respectively. We obtained excellent fittings (solid lines) with all measured experimental data (symbols) and extracted their in-plane anisotropic dielectric permittivity (Fig. 3g and Supplementary Note 2). It can be seen that these materials show two distinct bands (650–800 cm^{-1} and 850–1,000 cm^{-1}), designated as stretching vibrations of TeO_4 and MoO_4 groups, respectively. They manifest discernible shifts in the reflection spectra. These shifts signify modifications in phonons and RBs, predominantly linked to the stretching and bending modes of the Mo–O bonds within the MoO_4 group, which are notably impacted by the substituted atoms[43]. The shifting of phonon frequencies is influenced by both the mass and ionic radius of the substituted atoms[34,44]. Specifically, lighter substitute atoms (Cd > Mn ≈ Zn > Mg) induce faster vibrations and higher phonon frequencies, resulting in a blueshift in the reflection spectra. Furthermore, in the case of materials with similar atomic mass, such as $ZnTeMoO_6$ and $MnTeMoO_6$, the smaller ionic radius of Zn^{2+} accelerates bond vibrations relative to Mn^{2+}, leading to a blueshift in the reflection spectra[44].

To investigate the effects of atomic substitution on polariton behaviour, we conducted near-field imaging of anisotropic slab modes. Gold microdisks were fabricated on these flakes to act as infrared optical antennas[45] for launching the slab modes. The measured near-field images (Fig. 3c–f) reveal frequency-dependent wavefronts of the antenna-emitted polaritons. The observed wavefronts exhibit asymmetry, characterized by variations in fringe period and density, which can be attributed to the oblique illumination in the s-SNOM set-up[27,46]. It is important to note that although we showed the influence of atomic substitution on slab modes in Fig. 3, the edge modes were also observed in those compounds (Supplementary Fig. 11).

The variations in polariton features found in the different materials underscore the important influence of atomic substitutions on their behaviour (Fig. 3c–f). For $MgTeMoO_6$ (Fig. 3c), we observed a distinct topological transition of polariton wavefronts and dispersion (Supplementary Fig. 12) when increasing the frequency. At ω = 900 cm^{-1}, the wavefront patterns are hyperbolic, while at ω = 910 and 920 cm^{-1}, they became elliptical. In contrast, the other crystal flakes (Fig. 3d–f), like $ZnTeMoO_6$, mostly displayed elliptical wavefronts across all studied frequencies. $CdTeMoO_6$ consistently exhibits circular polaritonic patterns at these frequencies (Fig. 3f) due to its tetragonal unit cell with in-plane isotropy[11]. Intriguingly, both $MgTeMoO_6$ and $ZnTeMoO_6$ exhibit a change in the major axis of the elliptical wavefronts (indicated by black dashed elliptic lines), which changes from the [010] direction at ω = 910 cm^{-1} to the [100] direction at ω = 920 cm^{-1}. These trends can be explained by the change of anisotropic permittivity shown in Fig. 3g. We also observed less polaritonic fringes in the [010] direction compared to in the [100] direction due to higher propagation loss (Supplementary Fig. 13).

By analysing the polariton dispersion in the [100] direction (Fig. 3h), we observed that $CdTeMoO_6$ and $MnTeMoO_6$ exhibit higher

field confinement at the same frequency compared with $ZnTeMoO_6$ and $MgTeMoO_6$. Interestingly, all those materials displayed a rapid change in polaritonic wavelength within a narrow frequency band (~30 cm^{-1}), indicating an ultraslow group velocity v_g. For example, $CdTeMoO_6$ exhibits a group velocity of $v_g = 0.0004c$ at $\omega = 910$ cm^{-1}, while $MgTeMoO_6$ demonstrates $v_g = 0.0006c$ at $\omega = 923$ cm^{-1} along the [100] direction (Supplementary Fig. 14). These group velocities are one order of magnitude smaller than those in hBN[41] and comparable to those of α-MoO_3 polaritons[15], providing interesting potential for slow light applications such as enhanced light–matter interactions.

The lifetime of polaritons is a crucial aspect of their future applications because it serves as an indicator of the duration of the light–matter interaction. To determine the lifetimes, we analysed the near-field amplitude profiles along the [100] direction and obtained the decay lengths (L). By employing the equation $\tau = L/v_g$, we calculated the lifetimes of polaritons in these quaternary oxides, which were found to be long, exceeding 2 ps (Supplementary Fig. 15). For example, $MgTeMoO_6$ exhibits a lifetime of up to 4 ps, while $CdTeMoO_6$ displays a lifetime of 5 ps. The use of low-loss substrates such as CaF_2 and BaF_2 instead of the relatively lossy SiO_2 substrate can further improve these lifetimes by ~30% (ref. 41). In addition, similar to hBN[34] and α-MoO_3 (ref. 47), an isotopic enrichment process could offer the potential to further improve the polariton lifetime. The observed long lifetimes could be attributed to the low vibration damping of the strong Mo–O covalent bonds present in the tellurite molybdenum quaternary oxides.

Diffractionless polariton canalization

Another intriguing aspect is the natural overlap of the RBs between $ATeMoO_6$ quaternary oxides and α-MoO_3 (Fig. 3g). This correspondence provides ample elements to construct various heterostructures, enabling effective manipulation of polaritons (Supplementary Figs. 16 and 17). One example integrates α-MoO_3 and $CdTeMoO_6$ (Fig. 4a), combining in-plane hyperbolic and isotropic vdW polaritonic materials. Simulations (Fig. 4b,c) show that dipole-launched polaritons in this heterostructure propagate in a highly collimated way along the characteristic optical axis ([100], $\varepsilon_x < 0$) of α-MoO_3. Thus, the optical axis of α-MoO_3 can regulate such canalized polariton propagation (Supplementary Fig. 18), eliminating the need for precisely controlled twisting angles in double or triple α-MoO_3 heterostructures[23,48–51]. Figure 4d displays the Fourier transform of the simulated polariton fields, revealing that the canalization polaritons result from the extreme elliptical IFCs of the polariton wavevectors, which open in the k_y direction due to substantial damping. This canalization effect is universal for combinations of in-plane hyperbolic and isotropic vdW polaritonic materials, and is also achievable with, for example, $MgTeMoO_6$ and $CdTeMoO_6$, but is less pronounced in these materials due to larger optical absorptions in their overlapped RB range.

To experimentally validate this straightforward system of polariton canalization, we transferred an α-MoO_3 flake onto a $CdTeMoO_6$ flake (Fig. 4e) and placed a gold disk between the layers to excite polaritons. AFM topography showed surface protrusions (Fig. 4f) from the gold disk beneath. The near-field image (Fig. 4g) displayed efficient launch of canalization polaritons by the gold disk. These polaritons propagate along the $\varepsilon_x < 0$ direction of α-MoO_3 with patterns similar to the simulation in Fig. 4c. We examined the width of the polariton beam at two distances and found it unchanged (Fig. 4h), matching the width of the gold disk, which demonstrates polariton canalization for diffraction-free information transfer.

Conclusions

Our study showcases the versatility and rich phenomenology offered by vdW quaternary oxides, establishing them as a platform for investigating anisotropic and low-loss polariton features. We have demonstrated the existence of both in-plane anisotropic slab-confined and in-plane anisotropic edge-confined polaritons in these materials, expanding the family of anisotropic polaritons. Furthermore, the long lifetimes of polaritons in these materials highlight their potential for facilitating prolonged light–matter interactions, thus offering promising prospects for applications in nanophotonics and optoelectronics. Although our investigation focuses on TeMo-based compounds, the substitution of three elements in these quaternary oxides offers a diverse range of other possibilities, such as TeW-, SbMo- and SbW-based compounds. Venturing into uncharted research domains, such as topological phonons, should make it possible to unlock the potential for discovering exotic topological polariton behaviour within these systems. Furthermore, leveraging the strong optical nonlinearities of these quaternary oxides, including second-harmonic generation, opens exciting avenues for enhancing their functional capabilities in polariton modulation, polariton lasering and polaritonic information processing.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41565-024-01628-y.

References

1. Halasyamani, P. S. & Poeppelmeier, K. R. Noncentrosymmetric oxides. *Chem. Mater.* **10**, 2753–2769 (1998).
2. Ok, K. M., Chi, E. O. & Halasyamani, P. S. Bulk characterization methods for non-centrosymmetric materials: second-harmonic generation, piezoelectricity, pyroelectricity, and ferroelectricity. *Chem. Soc. Rev.* **35**, 710–717 (2006).
3. Zhao, S. et al. $ZnTeMoO_6$: a strong second-harmonic generation material originating from three types of asymmetric building units. *RSC Adv.* **3**, 14000–14006 (2013).
4. Ra, H.-S., Ok, K. M. & Halasyamani, P. S. Combining second-order Jahn–Teller distorted cations to create highly efficient SHG materials: synthesis, characterization, and NLO properties of $BaTeM_2O_9$ (M = Mo^{6+} or W^{6+}). *J. Am. Chem. Soc.* **125**, 7764–7765 (2003).
5. Zhang, J. et al. Top-seeded solution growth, morphology, and properties of a polar crystal $Cs_2TeMo_3O_{12}$. *Cryst. Growth Des.* **11**, 1863–1868 (2011).
6. Gao, Z., Tao, X., Yin, X., Zhang, W. & Jiang, M. Elastic, dielectric, and piezoelectric properties of $BaTeMo_2O_9$ single crystal. *Appl. Phys. Lett.* **93**, 252906 (2008).
7. Wu, Q. et al. Biaxial crystal β-$BaTeMo_2O_9$: theoretical analysis and the feasibility as high-efficiency acousto-optic Q-switch. *Opt. Express* **25**, 24893–24900 (2017).
8. Forzatti, P., Trifiro, F. & Villa, P. $CdTeMoO_6$, $CoTeMoO_6$, $MnTeMoO_6$, and $ZnTeMoO_6$: a new class of selective catalysts for allylic oxidation of butene and propylene. *J. Catal.* **55**, 52–57 (1978).
9. Guo, X., Gao, Z. & Tao, X. Recent advances in tellurite molybdates/tungstates crystals. *CrystEngComm* **24**, 7516–7529 (2022).
10. Xie, C., Yuan, H., Liu, Y. & Wang, X. Two-nodal surface phonons in solid-state materials. *Phys. Rev. B* **105**, 054307 (2022).
11. Li, C. et al. Controlled growth of layered acentric $CdTeMoO_6$ single crystals with linear and nonlinear optical properties. *Cryst. Growth Des.* **18**, 3376–3384 (2018).
12. Geim, A. K. & Grigorieva, I. V. Van der Waals heterostructures. *Nature* **499**, 419–425 (2013).
13. Basov, D., Fogler, M. & García de Abajo, F. Polaritons in van der Waals materials. *Science* **354**, aag1992 (2016).
14. Low, T. et al. Polaritons in layered two-dimensional materials. *Nat. Mater.* **16**, 182–194 (2017).
15. Ma, W. et al. In-plane anisotropic and ultra-low-loss polaritons in a natural van der Waals crystal. *Nature* **562**, 557–562 (2018).

16. Zheng, Z. et al. A mid-infrared biaxial hyperbolic van der Waals crystal. *Sci. Adv.* **5**, eaav8690 (2019).
17. Abedini Dereshgi, S. et al. Lithography-free IR polarization converters via orthogonal in-plane phonons in α-MoO_3 flakes. *Nat. Commun.* **11**, 5771 (2020).
18. Liu, Y., Chen, X. & Xu, Y. Topological phononics: from fundamental models to real materials. *Adv. Funct. Mater.* **30**, 1904784 (2020).
19. Taboada-Gutiérrez, J. et al. Broad spectral tuning of ultra-low-loss polaritons in a van der Waals crystal by intercalation. *Nat. Mater.* **19**, 964–968 (2020).
20. Zhang, X. et al. Ultrafast anisotropic dynamics of hyperbolic nanolight pulse propagation. *Sci. Adv.* **9**, eadi4407 (2023).
21. Dai, S. et al. Tunable phonon polaritons in atomically thin van der Waals crystals of boron nitride. *Science* **343**, 1125–1129 (2014).
22. Li, P. et al. Infrared hyperbolic metasurface based on nanostructured van der Waals materials. *Science* **359**, 892–896 (2018).
23. Hu, G. et al. Topological polaritons and photonic magic angles in twisted α-MoO_3 bilayers. *Nature* **582**, 209–213 (2020).
24. Feres, F. H. et al. Sub-diffractional cavity modes of terahertz hyperbolic phonon polaritons in tin oxide. *Nat. Commun.* **12**, 1995 (2021).
25. Ni, G. et al. Long-lived phonon polaritons in hyperbolic materials. *Nano Lett.* **21**, 5767–5773 (2021).
26. Hu, H. et al. Doping-driven topological polaritons in graphene/α-MoO_3 heterostructures. *Nat. Nanotechnol.* **17**, 940–946 (2022).
27. Ma, W. et al. Ghost hyperbolic surface polaritons in bulk anisotropic crystals. *Nature* **596**, 362–366 (2021).
28. Hu, C. et al. Source-configured symmetry-broken hyperbolic polaritons. *eLight* **3**, 14 (2023).
29. Hu, G. et al. Real-space nanoimaging of hyperbolic shear polaritons in a monoclinic crystal. *Nat. Nanotechnol.* **18**, 64–70 (2023).
30. Chaudhary, K. et al. Engineering phonon polaritons in van der Waals heterostructures to enhance in-plane optical anisotropy. *Sci. Adv.* **5**, eaau7171 (2019).
31. Chen, S. et al. Real-space observation of ultraconfined in-plane anisotropic acoustic terahertz plasmon polaritons. *Nat. Mater.* **22**, 860–866 (2023).
32. Qin, T., Ma, W., Wang, T. & Li, P. Phonon polaritons in van der Waals polar heterostructures for broadband strong light–matter interactions. *Nanoscale* **15**, 12000–12007 (2023).
33. Chen, J. et al. Optical nano-imaging of gate-tunable graphene plasmons. *Nature* **487**, 77–81 (2012).
34. Giles, A. J. et al. Ultralow-loss polaritons in isotopically pure boron nitride. *Nat. Mater.* **17**, 134–139 (2018).
35. Li, P. et al. Optical nanoimaging of hyperbolic surface polaritons at the edges of van der Waals materials. *Nano Lett.* **17**, 228–235 (2017).
36. Dai, S. et al. Manipulation and steering of hyperbolic surface polaritons in hexagonal boron nitride. *Adv. Mater.* **30**, 1706358 (2018).
37. Caldwell, J. D. et al. Sub-diffractional volume-confined polaritons in the natural hyperbolic material hexagonal boron nitride. *Nat. Commun.* **5**, 5221 (2014).
38. Li, P. et al. Hyperbolic phonon-polaritons in boron nitride for near-field optical imaging and focusing. *Nat. Commun.* **6**, 7507 (2015).
39. Álvarez-Pérez, G., Voronin, K. V., Volkov, V. S., Alonso-González, P. & Nikitin, A. Y. Analytical approximations for the dispersion of electromagnetic modes in slabs of biaxial crystals. *Phys. Rev. B* **100**, 235408 (2019).
40. Yang, X., Yao, J., Rho, J., Yin, X. & Zhang, X. Experimental realization of three-dimensional indefinite cavities at the nanoscale with anomalous scaling laws. *Nat. Photon.* **6**, 450–454 (2012).
41. Yoxall, E. et al. Direct observation of ultraslow hyperbolic polariton propagation with negative phase velocity. *Nat. Photon.* **9**, 674–678 (2015).
42. Jin, C. & Li, Z. Synthesis, crystal structure, optical property and theoretical studies of a noncentrosymmetric telluromolybdate $CoTeMoO_6$. *J. Alloy. Compd.* **722**, 381–386 (2017).
43. Mączka, M. et al. Growth and characterization of nonlinear optical telluromolybdate $CoTeMoO_6$ single crystals. *J. Solid State Chem.* **220**, 142–148 (2014).
44. Gupta, M., Rambadey, O. V. & Sagdeo, P. R. Probing the effect of R-cation radii on structural, vibrational, optical, and dielectric properties of rare earth (R=La, Pr, Nd) aluminates. *Ceram. Int.* **48**, 23072–23080 (2022).
45. Yao, Z. et al. Probing subwavelength in-plane anisotropy with antenna-assisted infrared nano-spectroscopy. *Nat. Commun.* **12**, 2649 (2021).
46. Huber, A., Ocelic, N., Kazantsev, D. & Hillenbrand, R. Near-field imaging of mid-infrared surface phonon polariton propagation. *Appl. Phys. Lett.* **87**, 081103 (2005).
47. Zhao, Y. et al. Ultralow-loss phonon polaritons in the isotope-enriched α-MoO_3. *Nano Lett.* **22**, 10208–10215 (2022).
48. Chen, M. et al. Configurable phonon polaritons in twisted α-MoO_3. *Nat. Mater.* **19**, 1307–1311 (2020).
49. Duan, J. et al. Twisted nano-optics: manipulating light at the nanoscale with twisted phonon polaritonic slabs. *Nano Lett.* **20**, 5323–5329 (2020).
50. Zheng, Z. et al. Phonon polaritons in twisted double-layers of hyperbolic van der Waals crystals. *Nano Lett.* **20**, 5301–5308 (2020).
51. Duan, J. et al. Multiple and spectrally robust photonic magic angles in reconfigurable α-MoO_3 trilayers. *Nat. Mater.* **22**, 867–872 (2023).

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Methods

Materials growth and sample preparation

Single-crystal samples of $ATeMoO_6$ (A = Zn, Mn, Co, Mg, Cd) were grown using a self-serve melt method as described in previous works[3,52]. To distinguish between these different materials, we employed Raman spectroscopy to analyse the plate-like crystals, as seen in Supplementary Note 1. The layered flakes are mechanically exfoliated onto Si/SiO_2 (300 nm) substrates using conventional scotch tape exfoliation. For nanoimaging of the polariton wavefronts, we used gold-antenna-based polariton launching. We carefully select large-sized flakes for further processing by an optical microscope to avoid the influence of polaritons reflected by the material edges in experiments. Subsequently, gold disks were fabricated via standard electron beam lithography. The disk patterns were written on the resist spin-coated on the selected flakes by employing a marker-based position. Electron-beam evaporation was subsequently used to deposit 10 nm of chromium and 40 nm of gold in a vacuum chamber to fabricate the gold disks. The standard lift-off procedure was used to remove the developed resist and residual organic material.

Numerical simulations

We used the finite-element-method software COMSOL Multiphysics for our full-wave simulations. In Figs. 2f and 4b,c, we combined frequency-domain study with boundary mode analysis and calculated the polariton mode's electric field distribution $\mathrm{Re}(E_{[100]})$ and $\mathrm{Re}(E_{[010]})$ along the propagation direction. For the false-colour map shown in Fig. 2h, we used the 4 × 4 transfer matrix method to calculate the reflection coefficient[53]. In all our simulations, the permittivity of $ATeMoO_6$ used in this work is calculated according to a Lorentz model (see Supplementary Note 2 for more details). Three Lorentz oscillator models were used to describe the infrared permittivity of $ATeMoO_6$:

$$\varepsilon_{x,y,z}(\omega) = \varepsilon_\infty + \sum_i \frac{f_i^2}{\omega_i^2 - \omega^2 - i\gamma_i\omega} \quad (1)$$

where ε_∞ is the high-frequency dielectric constant, and f_i, ω_i and γ_i are the strength, Lorentzian oscillation frequency and damping rate, respectively (i = 1, 2, 3). The permittivities of α-MoO_3 and α-V_2O_5 are obtained from refs. 19,54, respectively.

Near-field optical nanoimaging

The infrared nanoimaging of polaritons in vdW quaternary oxides is conducted following previous work[27]. The platinum-coated AFM tip oscillates vertically with an amplitude of ~80 nm at a frequency $\Omega \approx 270$ kHz. Illumination is facilitated by a wavelength-tunable continuous-wave quantum cascade laser (850–959 cm^{-1}). The backscattered light is collected with a pseudo-heterodyne interferometer, which demodulates the detector signal at a higher harmonic $n\Omega$ ($n \geq 2$), providing the background-free signal. This process yields near-field amplitude s_n and phase φ_n images, with our figures displaying the amplitude s_3 images.

FTIR spectroscopy

Mid-infrared reflectance measurements were conducted using a Bruker microscope, which was coupled to a Bruker LUMOS II FTIR spectrometer (Bruker Optics). The spectrometer was equipped with a broadband mercury cadmium telluride detector covering the range of 600–4,000 cm^{-1}. The reflection spectra were obtained by off-normal-incidence illumination (average incidence angle, 25°). To obtain polarized reflection spectra, an infrared polarizer was positioned within the internal optical path. The spectra were collected with a spectral resolution of 4 cm^{-1} and the measurements were performed using a 50 × 50 μm^2 aperture size. A gold film was used as a reference for all measurements, which were carried out at room temperature and ambient pressure. The polarized reflection spectrum was obtained by systematically rotating a polarizer in 10° steps, resulting in a series of spectra. By comparing these spectra, we were able to extract the reflection spectra for the two characteristic directions.

Topological phonon calculation

The topological phonon band structures of $ZnTeMoO_6$ in the Γ-XY plane were obtained through first-principles calculations with the commercial software CASTEP code in Materials Studio, as implemented in the Vienna Ab Initio Simulation Package. The projector augmented wave method and generalized gradient approximation with Perdew–Burke–Ernzerhof functions were used for geometry optimization, and to calculate the electronic properties and the exchange-correlation interaction. The plane-wave cut-off energy is set as 500 eV for the structural relaxation. The k-mesh sample used for $ZnTeMoO_6$ is 3 × 3 × 2 in the Brillouin zone. The phonon dispersions of the 2 × 2 × 1 $ZnTeMoO_6$ cell were examined with the finite-displacement method. We take 5 × 5 × 3 k-points to obtain the phonon frequency distribution and then calculate the density function and the PDOS.

Raman characterization

Raman spectra of $ATeMoO_6$ were collected using commercial confocal Raman microscopes (Alpha 300R Raman microscope, WITec). These measurements were performed with an excitation laser of 532 nm and 100× objective lens (numerical aperture, 0.9). The laser spot size at the sample surface was ~2 μm in diameter. An 1,800 lines mm^{-1} grating was utilized, and the laser power was set at ~5.0 mW.

Data availability

All data are available in the paper or the Supplementary Information.

References

52. Zhao, S. et al. A combination of multiple chromophores enhances second-harmonic generation in a nonpolar noncentrosymmetric oxide: $CdTeMoO_6$. *J. Mater. Chem. C* **1**, 2906–2912 (2013).
53. Passler, N. C. & Paarmann, A. Generalized 4×4 matrix formalism for light propagation in anisotropic stratified media: study of surface phonon polaritons in polar dielectric heterostructures. *J. Opt. Soc. Am. B* **34**, 2128–2139 (2017).
54. Álvarez-Pérez, G. et al. Infrared permittivity of the biaxial van der Waals semiconductor α-MoO_3 from near-and far-field correlative studies. *Adv. Mater.* **32**, 1908176 (2020).

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Author contributions

P.L. and S.Z. conceived the study. T.S. fabricated the samples and performed the s-SNOM measurements with the help of W.M. R.C. and T.S. carried out the fitting of the permittivity of the materials. H.W. synthesized the materials with the help of S.Z. T.S., R.C. and W.M. performed the simulations. T.S. and R.C. carried out the far-field experiments. Q.Y. performed the calculation of the phonon band structures. P.L., S.Z., X.Z. and J.L. coordinated and supervised the

work. T.S. and P.L. wrote the paper with input from all co-authors. All authors read and approved the final paper.

Competing interests

The authors declare no competing interests.

Additional information

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